BACHELOR OF COMPUTER APPLICATION

Machine Learning-Based Prediction of Academic Stress

Among Undergraduate Students in Nepal

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BCA Final-Year Research Project
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Agenda

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Research question

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Objectives

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Academic stress is measurable.

Student wellbeing can be examined through routinely collected demographic, academic, lifestyle, psychosocial, and stress-assessment information.

10

stress-assessment questions on a 1–5 scale

CENTRAL RESEARCH QUESTION

“Can routinely collected student information predict academic stress reliably, without learning patterns that fail outside the training sample?”

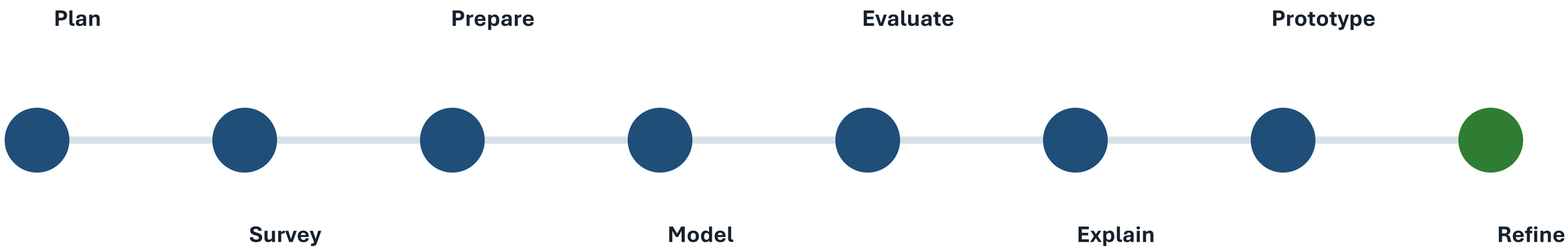
Objectives

Data Preparation

Model Training

Prototype & Insights

Development Methodology



16 WEEKS • AGILE / ITERATIVE

Data Sources and Stress Target Construction

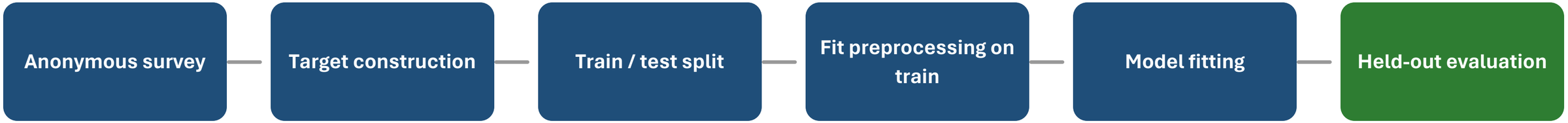
39	32	37	≈ 0.95
genuine responses	cleaned columns	final ML features	feature-to-sample ratio p/n

Anonymous Google Forms survey • voluntary • anonymous • about 5–7 minutes

Demographics | **Academic** | **Lifestyle** | **Psychosocial** | **10-question stress assessment**

stress_q1 to stress_q10 • 1–5 scale

Preprocessing Pipeline (Leakage-Safe)



The test set never informs feature preparation or model fitting.

07

Models Compared

Logistic Regression

K-Nearest Neighbors

Support Vector Machine

Decision Tree

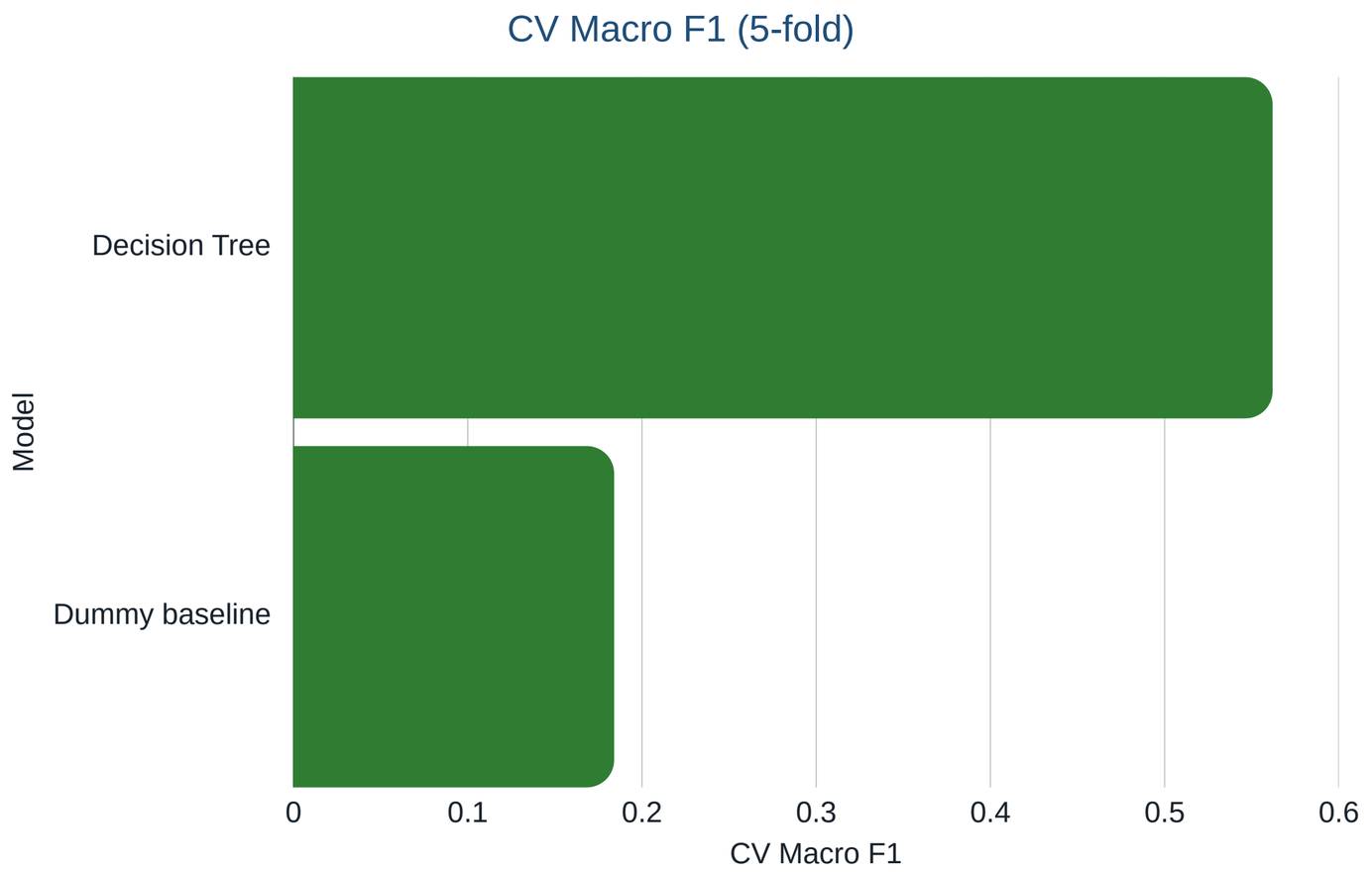
Random Forest

XGBoost

Naive Bayes

Dummy baseline

Real-Data Cross-Validation Results



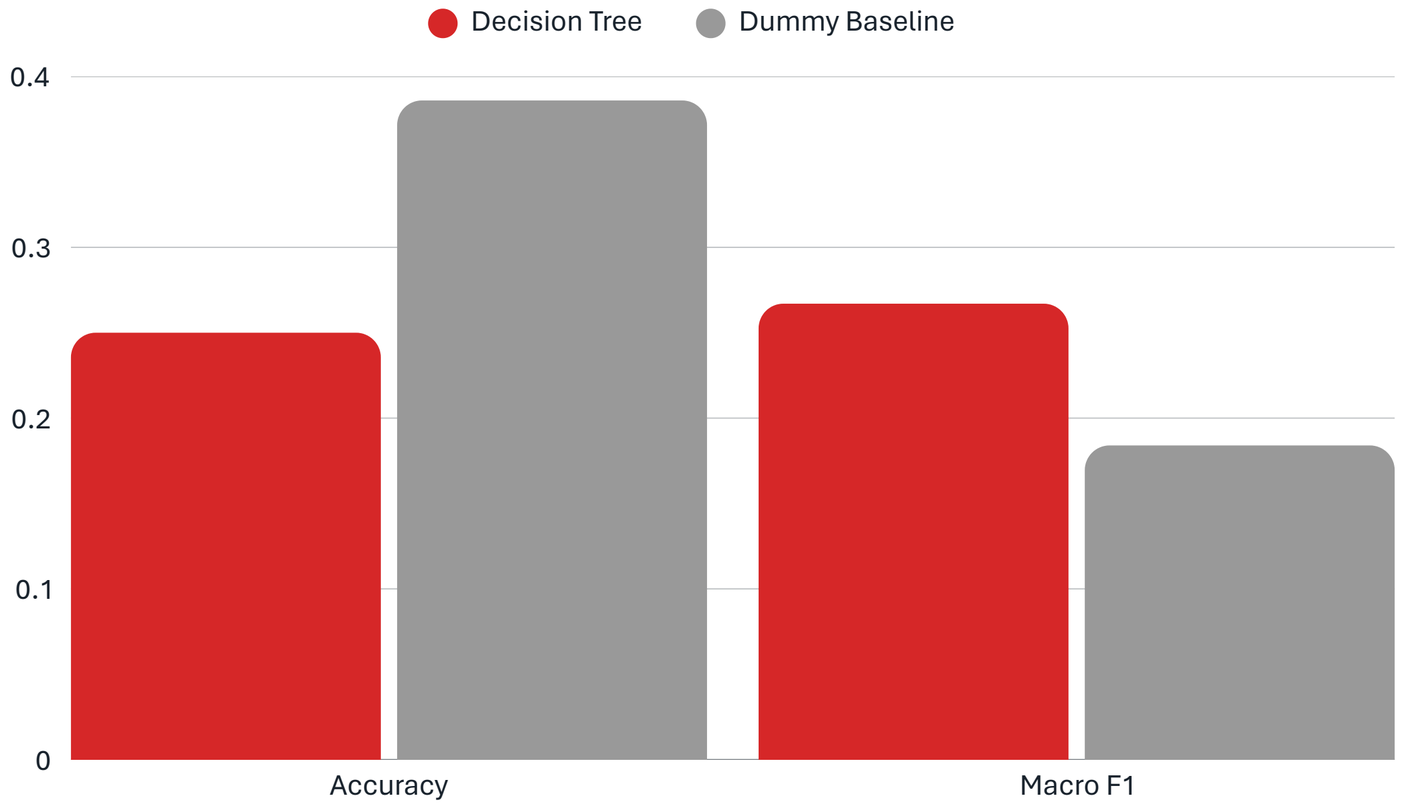
REAL DATA

n=39

Decision Tree
CV Macro F1 0.562

Only values reported in the study are shown.

Real-Data Generalization Gap



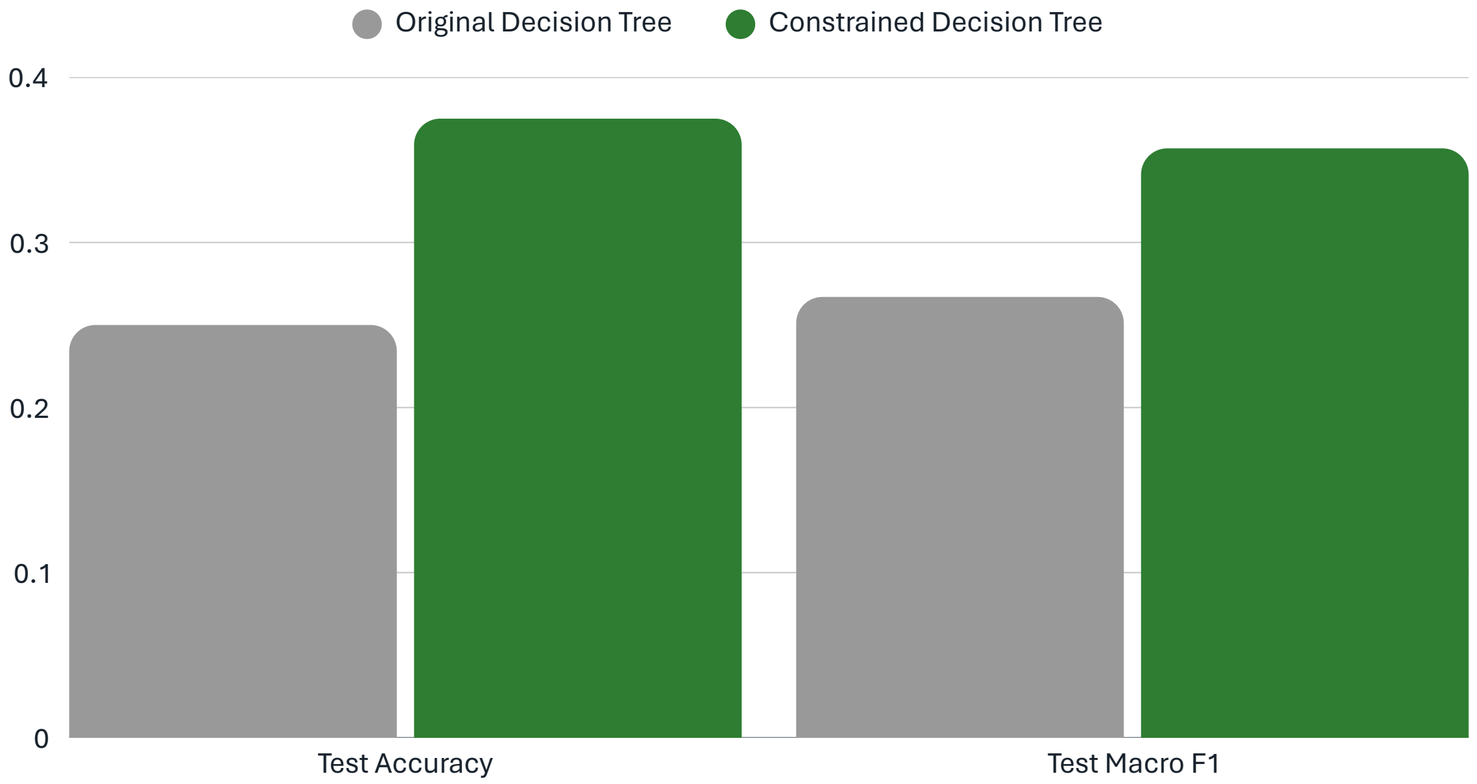
HONEST FINDING

0.562 → 0.267

CV Macro F1 to held-out Macro F1

The strongest cross-validation result did not generalize to the held-out sample.

Constrained Decision Tree Experiment



max_depth=3
min_samples_leaf=3

+0.125
accuracy gain

+0.090
Macro F1 gain

Synthetic Pipeline Validation (Validation Only)

LLM-GENERATED SYNTHETIC DATA

Methodological flaw — or insufficient sample size?

REAL
n=31
F1 0.267

SYNTHETIC
n=800

DELTA
reported comparison

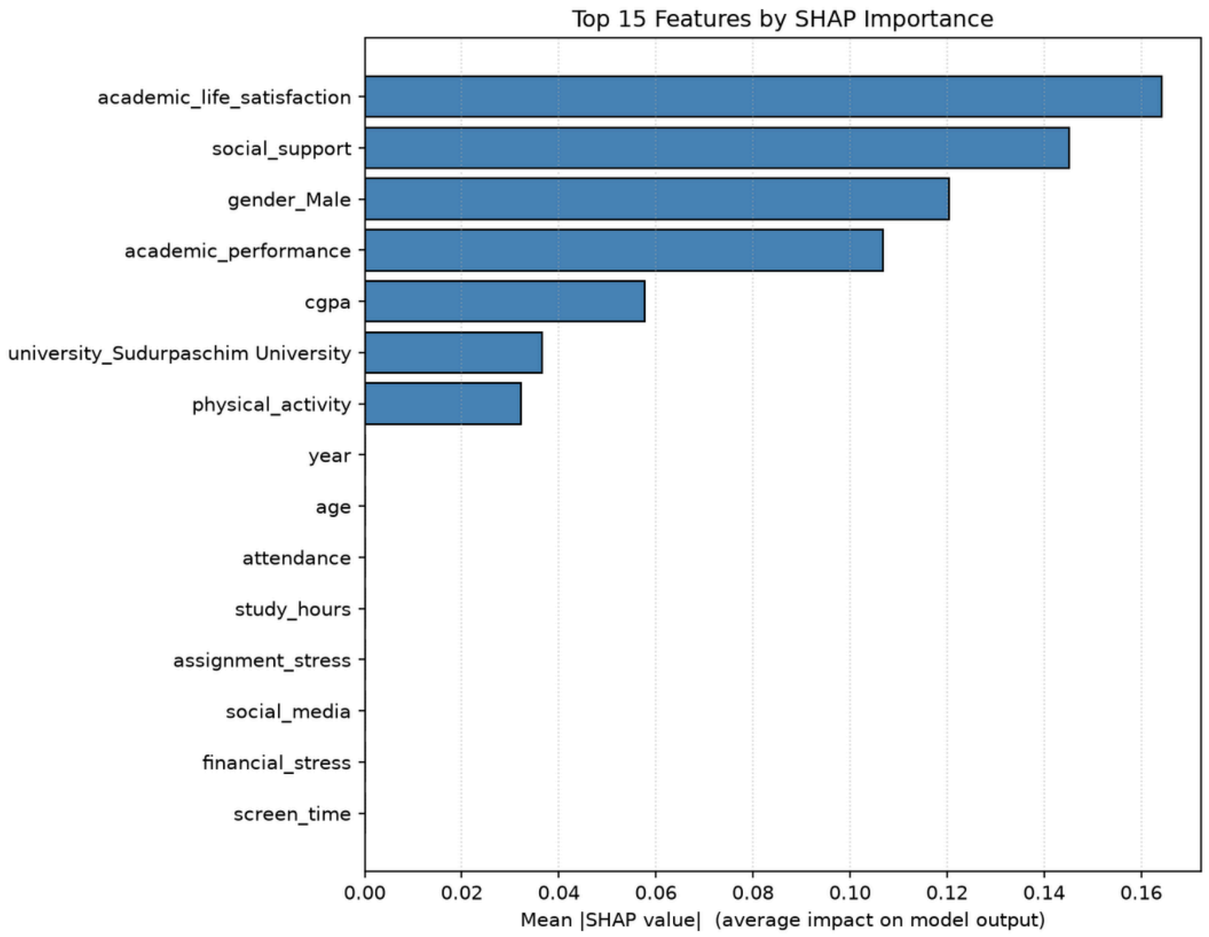
+0.34 mean F1(0.366 → 0.701)

Synthetic dataset: 1,000 rows • 35 columns • Data_Source = Synthetic
Classes: Low 332 • Moderate 342 • High 326 • Train 800 / Test 200
Synthetic best model: Random Forest

Explainability (SHAP) and Error Analysis

Top 7 features by mean absolute SHAP

Real data • Decision Tree



ERROR ANALYSIS

Small held-out sample

High feature-to-sample ratio

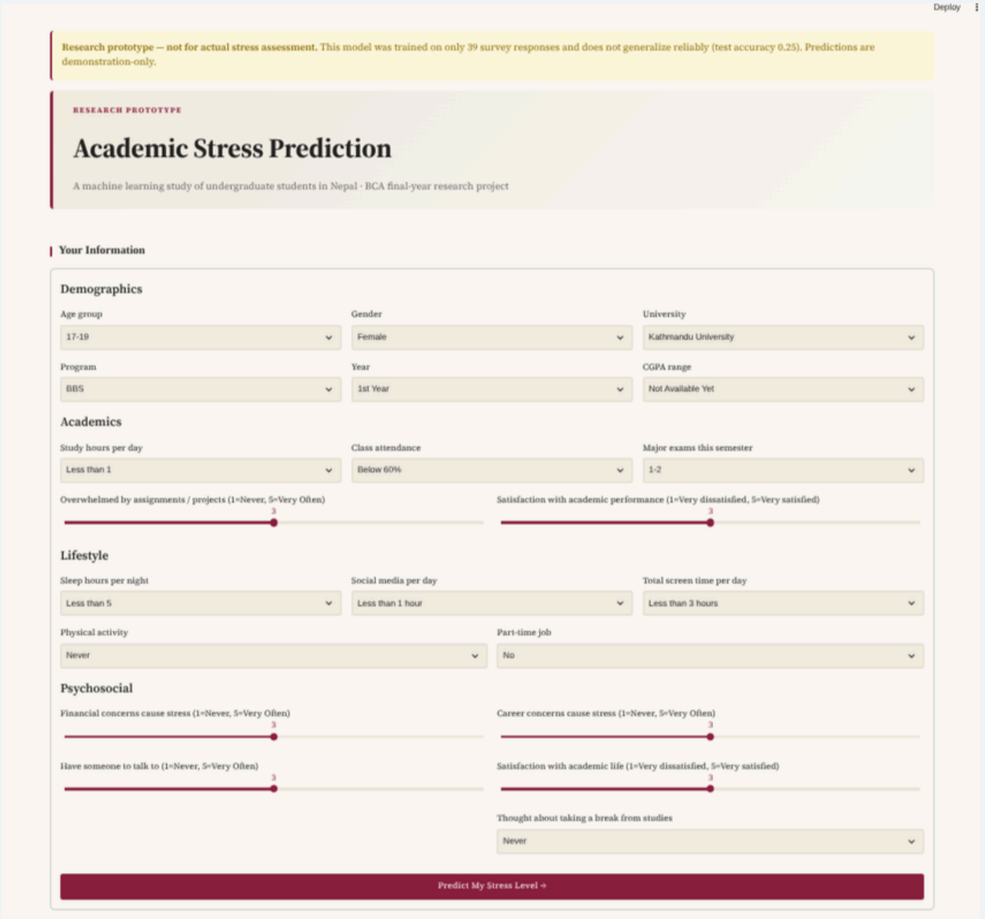
Cross-validation winner was unstable

The SHAP analysis identified academic life satisfaction, social support, and academic performance as the most influential features. Error analysis showed that all 6 held-out errors were between adjacent classes — never Low ↔ High. But the model was overconfident on wrong predictions, which is consistent with overfitting.

Streamlit Prototype

ACADEMIC STRESS PREDICTION

Student information
Academic factors
Lifestyle factors
Psychosocial factors



Conclusion and Future Work

CONCLUSION & LIMITATIONS

A leakage-safe pipeline was implemented, but the real dataset did not support reliable held-out generalization. Constraining the tree improved both test metrics.

- **End-to-end leakage-safe pipeline implemented**
- **Real-data CV F1 = 0.562; held-out F1 = 0.267**
- **Constrained tree (max_depth=3, min_samples_leaf=3) improved held-out F1 to 0.357**
- **Controlled synthetic experiment (n=1,000): mean F1 = 0.701**
- **Isolates sample size — not methodology — as the primary constraint**

FUTURE WORK

•Collect 300–500 genuine responses• Deploy the Streamlit prototype publicly to accelerate data collection via SQLite

• Re-run the pipeline once sufficient real data is collected

•Investigate feature reduction and regularization

•Perform external validation across institutions

THANK YOU

Questions & Discussion

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